UNIT 1 · L01

Data

UNIT 1 · L01

Information

UNIT 1 · L01

Bit

UNIT 1 · L01

Nibble

UNIT 1 · L01

Byte

UNIT 1 · L01

Kilobyte (KB)

UNIT 1 · L01

Megabyte (MB)

UNIT 1 · L01

Gigabyte (GB)

Backs · print double-sided, flip on long edge

Data that has been processed and given context so that it has meaning. Information	Raw facts and figures that have no meaning or context on their own. Data
A group of 4 bits (half a byte). One hexadecimal digit can be stored in a nibble. Nibble	A binary digit: the smallest unit of digital data, either a 0 or a 1 (off or on). Bit
1024 bytes – enough for a short plain-text file. Kilobyte (KB)	A group of 8 bits. One keyboard character (for example a letter, a digit or a punctuation mark) takes up 1 byte. Byte
1024 megabytes – around 20 minutes of HD video. Gigabyte (GB)	1024 kilobytes – around 1 minute of MP3 audio. Megabyte (MB)

UNIT 1 · L01

Terabyte (TB)

UNIT 1 · L01

ASCII

UNIT 1 · L01

Information system

UNIT 1 · L02

Data type

UNIT 1 · L02

Integer

UNIT 1 · L02

Real

UNIT 1 · L02

Date/time

UNIT 1 · L02

Character

Backs · print double-sided, flip on long edge

American Standard Code for Information Interchange: a character set that gives each letter, digit and punctuation mark a unique binary code.

ASCII

1024 gigabytes – the capacity of a typical laptop or external hard drive.

Terabyte (TB)

The kind of data an item is (such as integer, real, date/time, character or string), which decides how it is stored and processed.

Data type

Hardware and software working together to take data as input, process it and produce information as output.

Information system

Data type for numbers that can have a decimal or fractional part, positive or negative. Real numbers also include whole numbers.

Real

Data type for whole numbers, positive or negative, with no decimal or fractional part.

Integer

Data type for a single letter, digit or symbol, stored using its code from the character set.

Character

Data type that stores dates and/or times so that they can be sorted, calculated with and shown in different formats.

Date/time

UNIT 1 · L02

String / text

UNIT 1 · L02

Currency

UNIT 1 · L02

Boolean / Yes-No

UNIT 1 · L02

Type check

UNIT 1 · L03

Pixel

UNIT 1 · L03

Bitmap graphic

UNIT 1 · L03

Vector graphic

UNIT 1 · L03

Resolution

Backs · print double-sided, flip on long edge

A database data type for money values, shown with a currency symbol and two decimal places. Currency	Data type for a sequence of characters (letters, digits and symbols) of variable length. String / text
A validation check that makes sure the data entered is of the correct data type. Type check	Field type that can only hold one of two possible values, for example Yes or No (True or False). Boolean / Yes-No
An image made of a grid of pixels, each of which can be a different colour. The colour code of every pixel is stored, plus metadata such as the dimensions. Pixelation occurs when it is enlarged. Example file type: JPEG. Bitmap graphic	The smallest element of a digital picture that a screen can show or an editor can change. Every pixel holds one colour, recorded as a binary code. Pixel
The number of pixels in an image (width × height) or per inch. More pixels give a sharper image but a bigger file size. Resolution	An image stored as a list of objects (lines, curves and shapes) with properties such as position, size, line colour and fill colour. Can be enlarged without loss of quality. Example file type: SVG. Vector graphic

UNIT 1 · L03

Colour depth

UNIT 1 · L03

Pixelation

UNIT 1 · L03

PPI / DPI

UNIT 1 · L03

Metadata

UNIT 1 · L04

Analogue signal

UNIT 1 · L04

Analogue-to-digital converter (ADC)

UNIT 1 · L04

Sampling

UNIT 1 · L04

Sample rate

Backs · print double-sided, flip on long edge

The blocky appearance of a bitmap image when it is enlarged too much, because the individual pixels become visible.

Pixelation

The number of bits used to store the colour of each pixel. n bits give 2^n colours.

Colour depth

Data about the data, e.g. the width, height and colour depth of an image, stored so that the image can be rebuilt for display.

Metadata

Pixels per inch (screen) / dots per inch (printing): measures of resolution.

PPI / DPI

A device that converts an analogue signal, such as sound from a microphone, into digital (binary) data by sampling it at regular intervals.

Analogue-to-digital converter (ADC)

A signal that varies smoothly and continuously, representing a physical quantity such as a sound wave.

Analogue signal

The number of samples of a sound taken each second, measured in hertz (Hz). CD audio uses 44 100 Hz (44.1 kHz).

Sample rate

Measuring the level (amplitude) of an analogue signal at regular time intervals; each reading is rounded to a whole-number level and stored in binary.

Sampling

UNIT 1 · L04

Bit depth

UNIT 1 · L04

Bit rate

UNIT 1 · L04

Streaming

UNIT 1 · L04

Buffer / buffering

UNIT 1 · L04

Stereo

UNIT 1 · L05

Data portability

UNIT 1 · L05

File format

UNIT 1 · L05

Compression

Backs · print double-sided, flip on long edge

The number of bits of data processed for each second of sound, usually in kilobits per second; for one channel it equals sample rate × bit depth. Bit rate	How many bits are allocated to every sound sample. More bits give more possible levels, so each sample is recorded more accurately. Bit depth
An area of memory that temporarily stores the next part of a video or song before it is played, so playback continues if the connection slows or drops briefly. Buffer / buffering	Playing video or audio over the internet while it is still being received, without waiting to download and permanently store the whole file. Streaming
Being able to move data between computers, devices or software applications without typing it in again, usually by saving it in a standard file format. Data portability	Sound recorded on two channels (left and right); it needs twice as much data as the same sound in mono (one channel). Stereo
Reducing the size of a file so it needs less storage and can be uploaded or downloaded more quickly. Compression	The standard way data is organised inside a file, shown by its file extension (e.g. .jpg, .csv), which determines which programs can open it. File format

UNIT 1 · L05

Lossy compression

UNIT 1 · L05

Lossless compression

UNIT 1 · L05

CSV

UNIT 1 · L05

PDF

UNIT 1 · L05

JPEG

UNIT 1 · L05

TIFF

UNIT 1 · L05

WAV

UNIT 1 · L05

MIDI

Backs · print double-sided, flip on long edge

Compression that reduces file size without losing any data, so the file can be restored exactly as it was, e.g. zip, png. Lossless compression	Compression that reduces file size by permanently removing some data, so some quality is lost and the original cannot be fully restored, e.g. jpeg, mp3. Lossy compression
Portable Document Format, developed by Adobe: keeps a document's fonts, images and layout the same on any device; usually read-only. PDF	Comma-separated values: a text format in which each line is a record and the fields are separated by commas; used to move table data between spreadsheets and databases. CSV
Tagged Image File Format: stores high-quality bitmap images that can move between Windows and Apple computers; files are larger than jpegs. TIFF	Image format using lossy compression whose level the user can control; small files, widely used for photographs on web pages. JPEG
Musical Instrument Digital Interface: stores and exchanges musical instructions (notes, instruments, timing) between electronic instruments and computers, not recorded sound. MIDI	A standard uncompressed audio format; high quality but large files. WAV

UNIT 1 · L05

RTF

UNIT 1 · L06

System software

UNIT 1 · L06

Application software

UNIT 1 · L06

Operating system

UNIT 1 · L06

Boot-up sequence

UNIT 1 · L06

Real-time processing

UNIT 1 · L06

Batch processing

UNIT 1 · L06

Multi-user system

Backs · print double-sided, flip on long edge

Software that runs and manages the computer hardware and acts as the interface between the hardware and application programs, e.g. the operating system, utilities and device drivers.

System software

Rich Text Format: a text document format with basic formatting that can be opened by different word processors and operating systems.

RTF

System software that manages the computer's resources – allocating memory, storage and processing time – and lets the hardware and application software work together.

Operating system

Software designed to let the user carry out a specific task, such as word processing, spreadsheets or web browsing.

Application software

Each input is processed the moment it is entered, so the output is produced in time to affect the next input, e.g. ticket booking or air traffic control.

Real-time processing

The steps carried out when a computer is switched on: instructions in ROM check the hardware and load the operating system from storage into RAM.

Boot-up sequence

A system in which many users at different terminals share the processor of one powerful computer, each given a time slice in turn (round robin).

Multi-user system

Similar data is collected into batches over a period and processed together later, without human involvement, often overnight, e.g. bills and payroll.

Batch processing

UNIT 1 · L06

Time slice

UNIT 1 · L06

Master file

UNIT 1 · L06

Transaction file

UNIT 1 · L06

Packaged software

UNIT 1 · L06

Tailored software

UNIT 1 · L07

Utility software

UNIT 1 · L07

File fragmentation

UNIT 1 · L07

Disk defragmenter

Backs · print double-sided, flip on long edge

A permanent or semi-permanent file holding main records (e.g. customer details and reading history) that is updated when batch processing takes place.

Master file

A short period of processor time given to a task or user before the processor moves on to the next one.

Time slice

Off-the-shelf software made for many users with common needs; low cost and ready to use, e.g. Microsoft Word.

Packaged software

A temporary file holding the transactions collected for a batch (e.g. meter readings); it is used to update the master file.

Transaction file

A program that carries out one specific task to help the operating system maintain, protect or speed up the computer, e.g. defragmenter, task scheduler, backup, antivirus.

Utility software

Bespoke software developed for one organisation's specific needs; usually expensive because one customer bears the whole cost.

Tailored software

A utility that rearranges the fragments of files on a disk so each file is stored in continuous blocks and free space is grouped together, speeding up file access.

Disk defragmenter

When a file's data blocks are stored in separate, non-adjacent places on a disk because there is not enough continuous free space.

File fragmentation

UNIT 1 · L07

Task scheduler

UNIT 1 · L07

Round robin

UNIT 1 · L07

Backup

UNIT 1 · L07

Restore

UNIT 1 · L07

Full backup

UNIT 1 · L07

Incremental backup

UNIT 1 · L07

Virus

UNIT 1 · L07

Antivirus software

Backs · print double-sided, flip on long edge

A scheduling method in which each task or user is given a time slice of the processor in turn, returning to the first after the last. Round robin	A utility that shares processor time between tasks (e.g. round robin) and runs jobs such as backups, virus scans or updates automatically at set times. Task scheduler
Copying data back from a backup to return data or a system to an earlier state. Restore	A copy of data stored separately, preferably off-site, so it can be restored if the original is lost or damaged. Backup
A backup that copies only the data that has changed since the last backup; quicker and smaller than a full backup. Incremental backup	A backup in which all the data is copied; it takes a long time and needs a lot of storage. Full backup
Utility software that detects viruses by comparing files with a database of virus signatures, then removes or quarantines them; it must be updated regularly. Antivirus software	A harmful program that attaches itself to a file and spreads to other files and computers when the file is opened. Virus

UNIT 1 · L07

Quarantine

UNIT 1 · L08

Database

UNIT 1 · L08

Table

UNIT 1 · L08

Flat file

UNIT 1 · L08

Record

UNIT 1 · L08

Field

UNIT 1 · L08

Key field / primary key

UNIT 1 · L08

Composite key

Backs · print double-sided, flip on long edge

An organised, structured collection of related data that can be searched, sorted and presented, usually held in one or more tables.

Database

Isolating an infected file so it cannot run or spread, while keeping it in case it needs to be restored.

Quarantine

A database that holds all its data in a single two-dimensional table, which often leads to repeated data.

Flat file

A database structure that stores data about one type of thing, organised in rows (records) and columns (fields).

Table

A single item of data stored about every object or person, forming a column in a table and identified by its field name.

Field

All of the related data about one object or person, forming a complete row in a table. A record is made up of fields.

Record

A primary key made up of two or more fields that together uniquely identify a record.

Composite key

A field whose value is unique for every record, used to identify each record in a table.

Key field / primary key

UNIT 1 · L08

Foreign key

UNIT 1 · L08

Relationship

UNIT 1 · L08

Relational database

UNIT 1 · L08

One-to-many relationship

UNIT 1 · L08

Cardinality

UNIT 1 · L08

Crow's foot

UNIT 1 · L08

Referential integrity

UNIT 1 · L08

Form

Backs · print double-sided, flip on long edge

A link between two tables made through a common field (primary key and foreign key). It can be one-to-one, one-to-many or many-to-many. Relationship	The primary key of one table stored as a field in a second table, used to link the two tables. Foreign key
One record in a table links to many records in another table, e.g. one customer makes many bookings. One-to-many relationship	A database in which data is held in several linked tables. Relational database
A symbol drawn at the 'many' end of a relationship line, used instead of the infinity sign (∞). Crow's foot	How the records in two related tables relate to each other: one-to-one, one-to-many or many-to-many. Cardinality
An on-screen layout used to enter new records, edit existing records or view records in a database. Form	A rule applied to a relationship that keeps the linked tables consistent, e.g. a foreign key value must match an existing primary key. Referential integrity

UNIT 1 · L08

Report

UNIT 1 · L08

Macro

UNIT 1 · L08

Import

UNIT 1 · L09

Data type

UNIT 1 · L09

Validation

UNIT 1 · L09

Presence check

UNIT 1 · L09

Length check

UNIT 1 · L09

Type check

Backs · print double-sided, flip on long edge

A small stored program that carries out a repetitive task automatically, usually at the click of a button. Macro	A database tool that presents data from a table or query in a formatted, user-friendly layout for printing or display, often grouped and sorted. Report
The kind of data a field can hold, e.g. Short Text, Number, Currency or Date/Time. It controls what can be entered and what can be done with the data. Data type	To bring data into an application from another file, such as a csv file or spreadsheet, instead of retyping it. Import
A check that makes sure data has been entered in a required field, so the field cannot be left blank. Presence check	An automatic check made by the computer as data is input, to make sure the data is sensible, reasonable and inside allowed limits. Data that breaks the rules is rejected; validation cannot confirm the data is correct. Validation
A check that makes sure the data entered is of the correct data type, e.g. numeric, text or date. Type check	A check that makes sure the data entered has an exact number of characters, or is no shorter or longer than set limits. Length check

UNIT 1 · L09

Format check

UNIT 1 · L09

Range check

UNIT 1 · L09

Input mask

UNIT 1 · L09

Verification

UNIT 1 · L09

AutoNumber

UNIT 1 · L09

Currency

UNIT 1 · L09

Yes/No

UNIT 1 · L09

OLE Object

Backs · print double-sided, flip on long edge

A check that makes sure the data entered lies between a lower limit and an upper limit.

Range check

A check that makes sure the data entered matches a set pattern, e.g. a postcode or an email address containing @.

Format check

A check that data has been entered or copied accurately, e.g. by double entry or a visual check against the source.

Verification

A pattern set on a field that controls which characters can be typed in each position, used to carry out a format check.

Input mask

A data type for money values, stored and displayed with two decimal places.

Currency

A data type, often used for a key field, in which the computer creates a new sequential number automatically for each record.

AutoNumber

Object Linking and Embedding: a data type for a file or object created by software other than the database, e.g. a spreadsheet or word-processed document.

OLE Object

A data type that holds one of only two values (a Boolean), such as Yes/No or True/False.

Yes/No

UNIT 1 · L09

Attachment

UNIT 1 · L09

Hyperlink

UNIT 1 · L09

Calculated field

UNIT 1 · L09

Lookup Wizard

UNIT 1 · L10

Query

UNIT 1 · L10

Criteria

UNIT 1 · L10

Comparison operator

UNIT 1 · L10

Logical operator

Backs · print double-sided, flip on long edge

A data type that stores a link to a web address (URL) or an email address. Hyperlink	A data type that lets files, such as a digital image, be attached to a record. Attachment
Creates a list of values so that a value can be chosen from a drop-down list and entered into the field. Lookup Wizard	A data type that stores the result of a calculation using other fields in the same record. Calculated field
The conditions a record must meet to be included in the results of a query, e.g. Age >= 18. Criteria	A way of searching a database to extract the records that meet set criteria. It checks each record and lists the data that satisfies the criteria. Query
An operator such as AND, OR or BETWEEN that combines or tests conditions and gives a true or false (Boolean) result. Logical operator	A symbol used to compare a value with other data, giving true or false: < less than, <= less than or equal to, > greater than, >= greater than or equal to, = equal to, <> not equal to. Comparison operator

UNIT 1 · L10

BETWEEN

UNIT 1 · L10

Group by

UNIT 1 · L10

Big data

UNIT 1 · L10

Volume

UNIT 1 · L10

Velocity

UNIT 1 · L10

Variety

UNIT 1 · L10

Structured data

UNIT 1 · L10

Unstructured data

Backs · print double-sided, flip on long edge

Organises query or report results by the values of a field, e.g. grouping members by squad.

Group by

A logical operator that selects values within a range, including both end values.

BETWEEN

The very large amount of data stored by organisations, e.g. terabytes of social media posts each day.

Volume

Very large sets of data that could be mined for information but cannot be analysed by normal means because of their volume, velocity and variety.

Big data

The many different types and formats of data, such as text, images, video and sensor readings, which may be structured or unstructured.

Variety

The speed at which data is created and must be processed, often in real time.

Velocity

Data with no fixed format that cannot easily be processed with normal database tools, e.g. social media posts, emails, videos and essays.

Unstructured data

Data organised in a fixed format, usually tables of fields and records, so it is easy to search and analyse, e.g. dates, names, addresses, card numbers.

Structured data

UNIT 1 · L10

Data analytics

UNIT 1 · L10

Data mining

UNIT 1 · L11

Spreadsheet

UNIT 1 · L11

Cell range

UNIT 1 · L11

Label

UNIT 1 · L11

Value

UNIT 1 · L11

Formula

UNIT 1 · L11

Spreadsheet function

Backs · print double-sided, flip on long edge

Searching very large data sets with software to discover patterns and relationships that were not previously known. Data mining	Using algorithms, programs or artificial intelligence to examine data and find patterns, trends and relationships so that conclusions can be drawn. Data analytics
A block of adjacent cells referred to together, written first cell:last cell, e.g. B5:B12. Cell range	Application software that organises data in a grid of rows and columns so it can be calculated, analysed and presented. A spreadsheet file (workbook) contains one or more worksheets. Spreadsheet
A number entered into a spreadsheet so that calculations can be carried out, e.g. 5, 23. Value	Text used to identify the contents of a row or column, e.g. Name, Age. Label
A built-in formula that does a specific job, e.g. SUM, AVERAGE, MAX, MIN, COUNT, IF and VLOOKUP. Spreadsheet function	An instruction in a cell to carry out a calculation using operators and cell references, e.g. =A1+B1. It always starts with = and the cell shows the result, which updates when the referenced cells change. Formula

UNIT 1 · L11

SUM function

UNIT 1 · L11

AVERAGE function

UNIT 1 · L11

MAX function

UNIT 1 · L11

MIN function

UNIT 1 · L11

COUNT / COUNTIF

UNIT 1 · L11

Data types (spreadsheet formats)

UNIT 1 · L11

Replicating formulae

UNIT 1 · L11

Fill handle

Backs · print double-sided, flip on long edge

Calculates the arithmetic mean of the values in a range, e.g. =AVERAGE(A1:A10).

AVERAGE function

Adds all the numbers in a range, e.g. =SUM(A1:A10) adds A1 through to A10.

SUM function

Returns the lowest value in a range, e.g. =MIN(A1:A10).

MIN function

Returns the highest value in a range, e.g. =MAX(A1:A10).

MAX function

Formats applied to cell contents to show what kind of data is stored, e.g. Currency, Number, Text, Date/Time, Percentage.

Data types (spreadsheet formats)

COUNT counts the cells in a range that contain numbers; COUNTIF counts only the cells that meet a condition, e.g. =COUNTIF(C2:C20,"Yes").

COUNT / COUNTIF

The small square at the bottom right of a selected cell; dragging it replicates the cell's formula into neighbouring cells.

Fill handle

Copying a formula down or across cells (by dragging the fill handle or Fill Down) so the spreadsheet adjusts the references to match each row or column. Saves time and reduces errors.

Replicating formulae

UNIT 1 · L11

Relative cell reference

UNIT 1 · L11

Absolute cell reference

UNIT 1 · L11

Merge and centre

UNIT 1 · L11

Wrap text

UNIT 1 · L11

Headers and footers

UNIT 1 · L11

Print area

UNIT 1 · L12

IF function

UNIT 1 · L12

Nested IF

Backs · print double-sided, flip on long edge

A cell reference that stays the same when the formula is copied, replicated or dragged down, shown with $ signs, e.g. B9.

Absolute cell reference

A cell reference that changes when the formula is replicated, so it refers to the matching cell in the new row or column, e.g. B9.

Relative cell reference

Formatting that makes long text continue onto extra lines inside the cell instead of spilling into the next cell.

Wrap text

Combining two or more adjacent cells into a single cell and centring the contents across it.

Merge and centre

A selected range of cells that is the only part of the worksheet printed.

Print area

Text placed at the top (header) or bottom (footer) of every printed page, e.g. a title, file name, date or page number.

Headers and footers

An IF function placed inside another IF function so that more than two outcomes are possible, e.g. =IF(B2>=80,"Distinction",IF(B2>=65,"Merit","Fail")).

Nested IF

A function that tests a condition and returns one value if it is true and another if it is false, e.g. =IF(A1>60,"PASS","FAIL").

IF function

UNIT 1 · L12

VLOOKUP

UNIT 1 · L12

Lookup table

UNIT 1 · L12

Conditional formatting

UNIT 1 · L12

Data validation

UNIT 1 · L12

Error alert

UNIT 1 · L12

Template

UNIT 1 · L12

Importing data

UNIT 1 · L12

Data modelling / what-if

Backs · print double-sided, flip on long edge

The range of cells searched by VLOOKUP; the values to be matched are in its first column.

Lookup table

Vertical lookup: a function that searches for a value in the first column of a table and returns a value from a specified column in the same row. It updates automatically if the source data changes.

VLOOKUP

An automatic check when data is entered that it is sensible and within acceptable limits, e.g. a range check of 1 to 16. It cannot check the data is correct.

Data validation

Formatting (e.g. fill or font colour) that is applied automatically to cells only when a specified condition is met, e.g. hours worked over 40 highlighted.

Conditional formatting

A pre-designed spreadsheet with layout, formatting and formulae set up, used as the starting point for new documents with the same format.

Template

A message displayed by data validation when invalid data is entered, asking the user to re-enter valid data.

Error alert

Using a spreadsheet to represent a real situation and changing input values to predict outcomes and see the effect of the change, e.g. changing a price to see the effect on profit.

Data modelling / what-if

Bringing data into a spreadsheet from another application or file (e.g. a CSV file exported from a database) without retyping it.

Importing data

UNIT 1 · L12

Charts and graphs

UNIT 1 · L12

Macro

UNIT 1 · L13

CPU

UNIT 1 · L13

ALU

UNIT 1 · L13

Control unit

UNIT 1 · L13

Immediate-access store

UNIT 1 · L13

Register

UNIT 1 · L13

Program counter

Backs · print double-sided, flip on long edge

A small program that stores a sequence of actions so they can be carried out automatically with one click or shortcut, saving time on repeated tasks. Macro	Visual representations of data (e.g. column, line, pie) used to compare values, show trends or show parts of a whole; they can be copied into other software. Charts and graphs
Arithmetic logic unit: the part of the CPU that carries out arithmetic calculations and logical operations and comparisons. ALU	Central processing unit. A microprocessor whose purpose is to process data and carry out the instructions of programs, managing the functions of the computer. CPU
Main memory (RAM) holding the programs and data currently in use; each location has a unique address. Immediate-access store	The part of the CPU that fetches and decodes instructions and sends control signals to co-ordinate the other components. Control unit
Register holding the address of the next instruction to be fetched; incremented after each fetch. Program counter	A very small, very fast memory location inside the CPU used for a specific purpose. Register

UNIT 1 · L13

MAR

UNIT 1 · L13

MDR

UNIT 1 · L13

IAR

UNIT 1 · L13

Accumulator

UNIT 1 · L13

Fetch–execute cycle

UNIT 1 · L14

Clock speed

UNIT 1 · L14

Core

UNIT 1 · L14

Cache

Backs · print double-sided, flip on long edge

Memory data register: holds the data or instruction being transferred to or from memory. MDR	Memory address register: holds the address of the memory location being read from or written to. MAR
Register in the ALU that holds the result of a calculation before it is stored in memory. Accumulator	Instruction address register (CCEA term): holds the instruction currently being decoded and executed. IAR
The number of fetch–execute cycles a CPU can carry out per second, measured in hertz (e.g. 3 GHz = 3 billion cycles per second). Clock speed	The repeated process in which the CPU fetches an instruction from memory, decodes it and executes it. Fetch–execute cycle
Small, very fast memory on or near the CPU that holds frequently used instructions and data, so they need not be fetched from slower RAM. Cache	An independent processing unit inside a CPU; each core has a control unit, an ALU and registers, so it can process instructions at the same time as other cores. Core

UNIT 1 · L14

RAM

UNIT 1 · L14

ROM

UNIT 1 · L14

Volatile

UNIT 1 · L14

Non-volatile

UNIT 1 · L14

Boot program

UNIT 1 · L14

Virtual memory

UNIT 1 · L14

Multi-core processor

UNIT 1 · L15

Input device

Backs · print double-sided, flip on long edge

Read only memory: non-volatile memory whose contents cannot normally be changed, used to store the boot-up instructions (BIOS). ROM	Random access memory: volatile read/write main memory that holds the operating system, programs and data currently in use. RAM
Describes memory or storage that keeps its contents without power, such as ROM and SSDs. Non-volatile	Describes memory that loses its contents when the power is switched off, such as RAM and cache. Volatile
Part of a storage drive used as extra memory when RAM is full; it is much slower than RAM. Virtual memory	Instructions stored in ROM that run when the computer is switched on, checking hardware and loading the operating system into RAM. Boot program
Hardware used to enter data into a computer, such as a microphone, mouse or graphics digitiser. Input device	A CPU that contains two or more cores, so it can process several instructions in parallel. Multi-core processor

UNIT 1 · L15

Output device

UNIT 1 · L15

Graphics digitiser

UNIT 1 · L15

Touch screen

UNIT 1 · L15

Laser printer

UNIT 1 · L15

3D printer

UNIT 1 · L15

Hard disc drive (HDD)

UNIT 1 · L15

Solid state drive (SSD)

UNIT 1 · L15

Optical storage

Backs · print double-sided, flip on long edge

An input device made of a flat pad and a stylus, used to capture freehand drawings or signatures.

Graphics digitiser

Hardware used to present information from a computer, such as speakers or a printer.

Output device

An output device that uses a laser to form the page on a charged drum, which attracts toner that is fused onto paper; fast and high quality.

Laser printer

A display with a touch-sensitive surface that works as both an input and an output device.

Touch screen

Magnetic storage using spinning platters and moving read/write heads; large capacity at low cost per gigabyte.

Hard disc drive (HDD)

An output device that builds a solid object layer by layer from a computer-aided design (CAD) model.

3D printer

Storage read and written by a laser, such as Blu-ray, the HD storage media named in the specification.

Optical storage

Non-volatile flash storage with no moving parts: faster, more durable and lower power than an HDD, but more expensive per gigabyte.

Solid state drive (SSD)

UNIT 1 · L15

Flash memory

UNIT 1 · L15

Smart card

UNIT 1 · L16

LAN

UNIT 1 · L16

WAN

UNIT 1 · L16

Standalone computer

UNIT 1 · L16

Internet

UNIT 1 · L16

World Wide Web

UNIT 1 · L16

Intranet

Backs · print double-sided, flip on long edge

A plastic card with an embedded chip that stores a small amount of data, such as a chip-and-PIN bank card.

Smart card

Non-volatile electronic memory with no moving parts, used in SSDs, USB drives and memory cards.

Flash memory

Wide area network: a network covering a large geographical area that uses telecommunications links such as fibre optic cables or satellites.

WAN

Local area network: a network covering a small geographical area such as one building or site, usually owned by the organisation.

LAN

The global network of networks: the hardware and connections that carry data between computers worldwide.

Internet

A computer that is not connected to a network, so its files and devices cannot be shared.

Standalone computer

A private network owned by an organisation that uses internet technology but can only be accessed by authorised users who log on.

Intranet

A collection of linked web pages stored on web servers and accessed over the internet using a browser.

World Wide Web

UNIT 1 · L16

Internet of Things (IoT)

UNIT 1 · L16

NIC

UNIT 1 · L16

Switch

UNIT 1 · L16

Router

UNIT 1 · L16

Bus topology

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Terminator

UNIT 1 · L16

Star topology

UNIT 1 · L16

Ring topology

Backs · print double-sided, flip on long edge

Network interface card: a circuit board in a computer that lets it connect to a network and provides a port for the network cable. NIC	Everyday physical devices with sensors and connectivity that send and receive data over the internet, e.g. fitness watches and smart thermostats. Internet of Things (IoT)
A device that connects networks, such as a LAN and the internet, and forwards data along the best route using IP addresses. Router	A device that connects computers on a LAN and forwards data only to the intended computer. Switch
A device at each end of a bus backbone that absorbs the signal so it does not bounce back along the cable. Terminator	A layout where all devices are attached to a single backbone cable with a terminator at each end. Bus topology
A layout where each device connects to two others in a closed loop and data travels in one direction. Ring topology	A layout where each device is connected by its own cable to a central switch or server. Star topology

UNIT 1 · L17

Wi-Fi

UNIT 1 · L17

Bluetooth

UNIT 1 · L17

Optical fibre

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4G

UNIT 1 · L17

5G

UNIT 1 · L17

Bandwidth

UNIT 1 · L17

Latency

UNIT 1 · L17

Pairing

Backs · print double-sided, flip on long edge

Short-range, low-power wireless technology that connects two nearby devices directly after they have been paired. Bluetooth	Wireless technology that uses radio waves and a wireless router or access point to connect devices to a network without cables. Wi-Fi
Fourth-generation mobile technology giving high-speed wireless data through a network of masts. 4G	Cable made of thin glass strands that carries data as pulses of light, giving very high bandwidth and immunity to electrical interference. Optical fibre
The amount of data that can be transmitted in a given time, measured in bits per second. Bandwidth	Fifth-generation mobile technology with faster speeds, much lower latency and capacity for more devices than 4G, but needing more masts. 5G
Linking two Bluetooth devices so they recognise each other and can exchange data. Pairing	The delay between data being sent and being received. Latency

UNIT 1 · L17

Repeater

UNIT 1 · L17

Interference

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Cybercrime

UNIT 1 · L18

Hacking

UNIT 1 · L18

Phishing

UNIT 1 · L18

Denial of service

UNIT 1 · L18

Cyber stalking

UNIT 1 · L18

Cyber defamation

Backs · print double-sided, flip on long edge

Unwanted signals or obstacles that weaken or corrupt transmitted data. Interference	A device that boosts a signal so it can travel over longer distances. Repeater
Gaining unauthorised access to a computer system or its data, for example to steal, change or delete data. Hacking	Any crime committed using a computer, a network or the internet. Cybercrime
An attack that floods a server or website with requests so that it cannot respond to genuine users. Denial of service	Emails, texts or websites that pretend to come from a trusted organisation in order to trick people into giving personal details such as passwords or bank details. Phishing
Posting untrue statements online that harm the good name of a person or an organisation. Cyber defamation	Using technology to repeatedly harass, track or frighten a person. Cyber stalking

UNIT 1 · L18

Digital forgery

UNIT 1 · L18

Spamming

UNIT 1 · L18

Identity fraud / theft

UNIT 1 · L18

Malware

UNIT 1 · L18

Virus

UNIT 1 · L18

Worm

UNIT 1 · L18

Trojan horse

UNIT 1 · L18

Key logger

Backs · print double-sided, flip on long edge

Sending bulk unwanted messages, usually adverts or scams, to many people. Spamming	Deliberately and falsely altering digital content, such as photos or documents, to deceive. Digital forgery
Malicious software designed to damage or disrupt a computer system or to gain unauthorised access to it or its data. Malware	Using another person's personal details to pretend to be them, for example to obtain credit or goods. Identity fraud / theft
Self-replicating malware that spreads across a network by itself without needing a host file, using up bandwidth. Worm	Malware that attaches itself to a file or program and spreads when the file is run, copying itself and damaging or deleting data. Virus
Malware that secretly records the keys a user presses and sends the log to a criminal, revealing passwords and card details. Key logger	Malware disguised as useful or legitimate software so the user installs it; it may delete files or open a back door for hackers. Trojan horse

UNIT 1 · L18

Spyware

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Anti-virus software

UNIT 1 · L19

Encryption

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Decryption

UNIT 1 · L19

Firewall

UNIT 1 · L19

Access rights / levels of access

UNIT 1 · L19

Read only

UNIT 1 · L19

Strong password

Backs · print double-sided, flip on long edge

Software that scans for, quarantines and removes known malware; it must be updated regularly to recognise new threats. Anti-virus software	Software installed secretly that monitors a user's activity and sends the information to a third party. Spyware
Turning encrypted data back into readable form using the correct key. Decryption	Scrambling data using an algorithm and a key so that it cannot be understood by anyone who does not have the key. Encryption
Permissions linked to a user ID that control what each user can see and do, e.g. no access, read only, read and write. Access rights / levels of access	Hardware or software that monitors and filters data entering or leaving a network, blocking traffic that does not meet security rules. Firewall
A long password that mixes upper and lower case letters, digits and symbols and is not a dictionary word or personal detail. Strong password	An access level where a user can view data but cannot change it. Read only

UNIT 1 · L19

Two-factor authentication

UNIT 1 · L19

Backup

UNIT 1 · L19

Protocol

UNIT 1 · L19

FTP

UNIT 1 · L19

HTTP

UNIT 1 · L19

HTTPS

UNIT 1 · L20

Cloud computing

UNIT 1 · L20

Cloud storage

Backs · print double-sided, flip on long edge

A copy of data stored separately from the original so it can be restored if the original is lost or corrupted. Backup	A logon that needs two different forms of verification, such as a password and a code sent to a phone. Two-factor authentication
File Transfer Protocol: rules for uploading and downloading files between computers over a network. FTP	An agreed set of rules governing how data is transmitted between devices. Protocol
HyperText Transfer Protocol Secure: HTTP with encryption, so data sent between browser and server cannot be read if intercepted. HTTPS	HyperText Transfer Protocol: rules a browser uses to request web pages from a web server and transfer them. HTTP
Keeping files on a provider's remote servers so they can be accessed from any device with an internet connection. Cloud storage	Storing data and running software on remote servers that are accessed on demand over the internet, instead of on a local computer. Cloud computing

UNIT 1 · L20

Data centre

UNIT 1 · L20

Scalability

UNIT 1 · L20

Pay as you go

UNIT 1 · L20

Downtime

UNIT 1 · L20

Cloud gaming

UNIT 1 · L20

Synchronisation

UNIT 1 · L20

Collaborative tool

UNIT 1 · L20

Version history

Backs · print double-sided, flip on long edge

The ability to increase or decrease storage, bandwidth or users quickly as an organisation's needs change. Scalability	A building containing large numbers of servers that store data and run cloud services. Data centre
A period when a service is unavailable, for example because of an internet failure or provider maintenance. Downtime	A payment model where an organisation pays only for the cloud resources it actually uses. Pay as you go
Keeping copies of a file up to date so that a change made in one place appears on every device and for every user. Synchronisation	Games run on remote servers and the video is streamed to the player's device over the internet. Cloud gaming
A record of the changes made to a file that shows who made them and allows earlier versions to be restored. Version history	Cloud software that lets several people view and edit the same file at the same time, e.g. Google Docs or Microsoft 365. Collaborative tool

UNIT 1 · L20

Carbon footprint

UNIT 1 · L21

Consumer Contracts Regulations 2013

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Cooling-off period

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Copyright, Designs and Patents Act 1988

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Intellectual property

UNIT 1 · L21

Software licence

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Software piracy

UNIT 1 · L21

Computer Misuse Act 1990

Backs · print double-sided, flip on long edge

The Consumer Contracts (Information, Cancellation and Additional Charges) Regulations 2013: protect people who buy online, by phone or by mail order by requiring clear information, giving a 14-day right to cancel and banning hidden extra charges.

Consumer Contracts Regulations 2013

A measure of the impact of an organisation's or person's activities on the environment.

Carbon footprint

Protects the intellectual property of creators, including software, music, images and text, so it cannot be copied, shared, sold or adapted without the owner's permission.

Copyright, Designs and Patents Act 1988

The 14 days in which a consumer can cancel a distance purchase without giving a reason and get a refund; for goods it starts the day after delivery.

Cooling-off period

A legal agreement that states how a piece of software may be used, including how many users or computers may run it.

Software licence

Original creative work, such as software, music, designs or writing, which belongs to the person or organisation that created it.

Intellectual property

Makes it a criminal offence to access computer material without permission, to do so in order to commit another crime, or to carry out unauthorised acts that impair a computer, such as changing or deleting data.

Computer Misuse Act 1990

Illegally copying, downloading, distributing or selling software without the copyright owner's permission.

Software piracy

UNIT 1 · L21

Unauthorised access

UNIT 1 · L21

Hacker

UNIT 1 · L21

Virus

UNIT 1 · L21

Spyware

UNIT 1 · L22

UK GDPR

UNIT 1 · L22

Data Protection Act 2018

UNIT 1 · L22

Personal data

UNIT 1 · L22

Special category data

Backs · print double-sided, flip on long edge

Someone who gets into a computer system, or the data held on it, without permission.

Hacker

Getting into a computer system, account or data without permission; an offence under section 1 of the Computer Misuse Act 1990.

Unauthorised access

Malicious software that secretly records a user's activity, such as keystrokes, and sends it to a third party.

Spyware

Malicious software that attaches itself to files or programs and copies itself, often damaging or deleting data.

Virus

UK law that works alongside the UK GDPR to control how personal data is collected, stored and used; it replaced the Data Protection Act 1998.

Data Protection Act 2018

The UK version of the General Data Protection Regulation, which, with the Data Protection Act 2018, sets the rules for processing personal data in the UK.

UK GDPR

Personal data that needs extra protection because its misuse could cause discrimination or harm, e.g. health, religion or ethnic origin.

Special category data

Any information about a living person who can be identified from it, such as their name, address, photo or email address.

Personal data

UNIT 1 · L22

Data subject

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Data controller

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Data Protection Officer

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Information Commission

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Subject access request

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Data breach

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Phishing

UNIT 1 · L23

Acceptable use policy

Backs · print double-sided, flip on long edge

The person or organisation that decides why and how personal data is processed and is responsible for following data protection law. Data controller	The living individual whom the personal data is about. Data subject
The independent UK regulator that enforces data protection law, investigates complaints and can issue fines; it replaced the Information Commissioner on 30 September 2026. Information Commission	A person who advises an organisation on data protection law, monitors that it is followed and acts as the contact for the regulator; required for public bodies such as schools. Data Protection Officer
A security incident in which personal data is lost, destroyed, changed or disclosed without authorisation. Data breach	A request by a data subject to see a copy of the personal data an organisation holds about them; normally answered free within one month. Subject access request
Rules that employees or pupils agree to that set out how they may use an organisation's computers and internet. Acceptable use policy	Using fake emails, texts or adverts that look official to trick people into giving away personal details such as passwords. Phishing

UNIT 1 · L23

Trolling

UNIT 1 · L23

Cyberbullying

UNIT 1 · L23

Fake news

UNIT 1 · L23

GPS

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Job displacement

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Automation

UNIT 1 · L23

Upskilling

UNIT 1 · L23

Teleworking

Backs · print double-sided, flip on long edge

Using technology, such as social media or messages, to repeatedly hurt, threaten or embarrass a particular person. Cyberbullying	Deliberately posting provocative or offensive messages online to upset people or start arguments; credited separately from cyberbullying. Trolling
Global Positioning System: uses signals from satellites to work out the exact location of a device in real time. GPS	Deliberately false or misleading information that is spread online, often made to look like real news. Fake news
Machines or software carrying out tasks that were previously done by people. Automation	When jobs are lost or changed because digital technology now does work that people used to do. Job displacement
Using ICT, such as broadband, laptops and video conferencing, to work from home instead of at the workplace. Teleworking	Training workers in new skills, especially digital skills, so they can keep up with changes in technology. Upskilling

UNIT 1 · L23

Repetitive strain injury (RSI)

UNIT 1 · L23

Ergonomic

UNIT 1 · L24

E-commerce

UNIT 1 · L24

Online banking

UNIT 1 · L24

Online training

UNIT 1 · L24

Gaming application

UNIT 1 · L24

Gamification

UNIT 1 · L24

Simulation

Backs · print double-sided, flip on long edge

Designed to fit the user and reduce physical strain, e.g. adjustable chairs, ergonomic keyboards and wrist rests. Ergonomic	Pain or injury in the hands, wrists, arms or shoulders caused by repeating the same movements, such as typing, for long periods. Repetitive strain injury (RSI)
Using a bank's secure website or app to manage an account, for example viewing statements, paying bills and transferring money. Online banking	The buying and selling of goods and services over the internet. E-commerce
An electronic game played on a console, computer or mobile device, often online with other players. Gaming application	Learning delivered over the internet, often with multimedia and automatic progress tracking, that learners can complete at their own pace. Online training
A computer model, built from data and mathematical rules, that copies how a real object or situation behaves so it can be tested or practised safely. Simulation	Adding game features such as points, badges and leaderboards to a non-game activity to increase motivation. Gamification

UNIT 1 · L24

Virtual reality

UNIT 1 · L24

Mobile app

UNIT 1 · L24

Phishing

UNIT 1 · L24

Two-factor authentication

UNIT 1 · L24

GUI

UNIT 1 · L24

Quality of written communication

Backs · print double-sided, flip on long edge

A software application designed to run on a smartphone or tablet, usually downloaded from an app store. Mobile app	A computer-generated 3D environment that the user enters using a headset and sometimes gloves or controllers, stimulating senses such as sight, hearing and touch. Virtual reality
Checking a user's identity using two different methods, such as a password plus a code sent to their phone. Two-factor authentication	Fake emails, texts or websites that pretend to be from a trusted organisation in order to trick people into giving personal or login details. Phishing
Credit in extended answers for organised writing, correct spelling and grammar, and accurate use of specialist terms. Quality of written communication	Graphical user interface: lets users interact using windows, icons, menus and a pointer, or touch on a mobile app. GUI